EMMANUEL TIGOUE, CISSP

AI Security Engineer | Security Engineering & Agentic Systems

Atlanta, GA | 404-839-2214 | etigoue@tigouetheory.com | LinkedIn | GitHub | Portfolio | U.S. Citizen

CERTIFICATIONS

CISSP | **SecurityX** | **CCNA** | **Security+** | AWS Security Specialty (in progress)

EXPERIENCE

AI Security Engineer | CoreDirective

Atlanta, GA | Sep 2025 - Present

- Architected Squire, an AI SOC analyst that drafts a triage verdict for every alert and cuts manual triage time per alert **80%**. Built on LangGraph with human approval gates, pgvector RAG, NeMo Guardrails PII redaction, and Langfuse tracing.
- Threat modeled the OpenClaw gateway (Claude Fable 5) and its MCP tool permissions against the OWASP LLM Top 10 and MITRE ATLAS, then set the **guardrail and access requirements the platform shipped with**.
- Ran structured red team assessments on production AI stacks and web applications: prompt injection, jailbreaks, data leakage, excessive agency, and OWASP web risks. Remediated all **6 high and medium findings** in **5 attack classes** before launch.
- Built the Open Policy Agent (OPA) gate for Knight, a red team agent for authorized testing: a Rego allowlist that checks scope authorization, exclusions, decoys, time window, kill switch, and signed approvals, with **100+ tests** at **100% coverage** in CI.
- Embedded Semgrep SAST, Trivy container image and dependency scanning, Gitleaks secrets scanning, and OPA policy gates for infrastructure as code into CI/CD. Added OWASP ZAP DAST against staging and Cosign image signing, with deploys rejecting unsigned images.
- Placed every admin surface behind Cloudflare Zero Trust, defined in Terraform with **zero public ports** and separate policies for human and service accounts.
- Tuned Falco eBPF container runtime detection shipped to Datadog and mapped to MITRE ATT&CK, cutting daily alerts from **200+ to 12** without losing coverage.
- Authored 57 GRC documents mapping **133 NIST 800-53 controls** plus NIST AI RMF, SOC 2, and ISO 27001, and built a Promptfoo eval harness that regression tests the AI guardrail controls.

IT Security & Operations Manager | Texaco

Atlanta, GA | Mar 2022 - Mar 2026

- Deployed Splunk Enterprise SIEM across **3** sites, centralizing FortiGate, Cisco switch, and Windows logs with on call alerting and a one year AWS S3 archive under Object Lock. Detection time fell from **48 hours to under 4**.
- Owned incident response for **45** endpoints, covering card skimming, credential compromise, and unauthorized vendor access. Wrote a **6** step IR runbook that reduced containment from **8 hours to 90 minutes**.
- Traced **$18K a year** in card fraud to skimming on outdated, tamper prone payment terminals and drove their replacement with current models on automatic vendor updates.
- Segmented a flat network into **6** VLANs behind FortiGate zones with default deny, isolating payment systems and tank gauges. Verified with Nmap and closed **12 of 14** critical PCI DSS gaps.
- Built a quarterly Nessus vulnerability management cycle with patch reporting automated in Python and PowerShell, and brought critical patching **from 90 to 30 days**.
- Removed standing admin rights and hardened Active Directory and Entra ID with MFA, FIDO2 admin keys, GPO baselines, stale account cleanup, and credential rotation.

TECHNICAL SKILLS

AI Security: LLM guardrail design, AI threat modeling, agentic architecture review, OWASP LLM Top 10, Model Context Protocol (MCP)

Cloud Security & Automation: AWS, Oracle Cloud, Docker, HashiCorp Vault, OPA policy gates, n8n, Ollama

Identity & Access Management (IAM): Teleport JIT PAM, Keycloak SSO, RBAC, Azure AD, least privilege design, mTLS

Frameworks: NIST AI RMF, NIST 800-53, ISO 42001, SOC 2, ISO 27001, PCI DSS

EDUCATION

Georgia State University, Andrew Young School of Policy Studies

B.A. in Economics

A.S. in Business Administration | Dean's List